

TIRTH SAKARIYA

CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation

 Parul University

Document Details

Submission ID

trn:oid::3117:619441689

Submission Date

Sep 2, 2026, 7:43 AM GMT+5:30

Download Date

Sep 2, 2026, 7:44 AM GMT+5:30

File Name

chronus (1).pdf

File Size

231.9 KB

10 Pages

5,647 Words

36,912 Characters

2% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Abstract
- Methods and Materials
- Preprints

Exclusions

- 5 Excluded Preprints

Match Groups

-  **6 Not Cited or Quoted 2%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 2%  Internet sources
- 1%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **6 Not Cited or Quoted** 2%
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations** 0%
Matches that are still very similar to source material
-  **0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 2%  Internet sources
- 1%  Publications
- 0%  Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

- 1** **Publication**
Keyur Mistry, Daxa Vekariya, Ankita Gandhi, Vipul Vekariya, Mukesh Patidar. "A R... 1%
- 2** **Internet**
sagarsamtani.com <1%
- 3** **Internet**
journal.kasell.or.kr <1%
- 4** **Internet**
www.ijert.org <1%
- 5** **Publication**
Nguyen, Vy Thuy. "Balancing Relevance and Diversity in Music Recommendation ... <1%
- 6** **Internet**
assets-eu.researchsquare.com <1%

CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation

Tirth Sakariya

Dept. of Computer Science and Engineering
Parul Institute of Engineering and Technology
Parul University, Vadodara, India
2303031050723@paruluniversity.ac.in

Manav Aghera

Dept. of Computer Science and Engineering
Parul Institute of Engineering and Technology
Parul University, Vadodara, India
2403031057168@paruluniversity.ac.in

Manish Prajapati

Dept. of Computer Science and Engineering
Parul Institute of Engineering and Technology
Parul University, Vadodara, India
2303031050721@paruluniversity.ac.in

Abstract—Human identity is encoded across fragmented personal archives—diaries, correspondence, interviews, digital communications, and voice recordings—yet these records remain static and non-interactive after a person’s death or incapacitation. Current large language models can generate plausible persona-styled text but frequently fabricate undocumented statements, lack provenance mechanisms, and offer no principled governance for sensitive personal data. This paper presents CHRONUS, a memory-grounded persona simulation system that transforms personal documentary evidence into an interactive, retrieval-based conversational representation. The architecture employs sentence-level semantic embedding via Sentence-BERT (all-MiniLM-L6-v2, 384 dimensions), locally persistent vector storage in ChromaDB, cosine-similarity retrieval with empirically calibrated confidence thresholds, and a structured three-part response pipeline that anchors generated output to retrieved source passages. A 25-question structured interview protocol augments sparse documentary corpora across six personality dimensions. Preliminary evaluation on an internal corpus demonstrates that semantic retrieval achieves Precision@1 of 0.82 compared with 0.41 for keyword-based matching, and that retrieval-grounded response construction produces substantially higher groundedness scores (4.6/5.0 vs. 2.3/5.0) than unconstrained persona prompting. The system operates on a local-first privacy architecture with no mandatory external data transmission for core functionality. We discuss ethical requirements for consent, posthumous governance, disclosure, and emotional safety, and identify limitations including small evaluation scale, absence of longitudinal user studies, and the fundamental gap between documented personality and lived human identity. CHRONUS contributes a practical framework for evidence-grounded personality simulation that prioritises provenance over imitation and documented memory over fabricated continuity.

Index Terms—Memory-grounded AI, persona simulation, retrieval-augmented generation, semantic memory, digital legacy, grief technology, voice synthesis, privacy-preserving AI, agentic architecture, conversational AI

I. INTRODUCTION

Human personality, knowledge, and identity exist across fragmented and heterogeneous records. A person’s documented life—letters, journals, interviews, emails, voice messages, and digital correspondence—constitutes a rich but structurally disorganised archive of their beliefs, values, communication patterns, and experiential knowledge. Upon death or cognitive decline, this archive becomes permanently static: searchable by keyword but fundamentally non-interactive, unable to respond to questions that were never explicitly addressed in the original material.

The emergence of large language models (LLMs) has created new possibilities for interactive personal representation. Systems such as GPT-4 [1] can generate convincing persona-styled language when conditioned on biographical descriptions, and commercial platforms like Replika [2] and Character.AI have demonstrated public appetite for AI companions that simulate distinct personalities. These systems establish that statistical language modelling can approximate communicative style with sufficient training data.

However, persona prompting and generative companionship introduce a fundamental integrity problem when applied to specific real individuals. LLMs generate text by predicting statistically probable continuations rather than retrieving verified personal information. When prompted to speak as a historical figure or deceased person, they routinely fabricate statements, opinions, and biographical details that the individual never expressed [1], [3]. For applications involving real human identity—particularly in contexts of bereavement, family memory, or historical preservation—such fabrication constitutes not merely an inaccuracy but a misrepresentation of a person’s documented record.

This integrity problem acquires particular urgency in the

domain of digital legacy and grief technology. Research on “continuing bonds” in bereavement psychology has established that maintaining symbolic connections with deceased loved ones can support healthy grieving [4], [5]. Yet commercial grief-tech platforms operate under significant tensions: the desire to create immersive, lifelike interactions may incentivise unconstrained generation at the expense of fidelity to the person’s actual documented expressions [6], [7]. Voinea [7] identifies the risk that griefbots may disrupt the natural grieving process by creating an “illusion of ongoing connection” that impedes the cognitive updating required for healthy adaptation to loss. Sri Takshara and Bhuvaneswari [8] extend this analysis to neurocognitive implications, arguing that AI-simulated presence may interfere with neuroplasticity-mediated emotional regulation.

CHRONUS addresses this constellation of challenges through a memory-grounded architecture that treats personal documentary evidence as the authoritative source for conversational responses. Rather than training or fine-tuning a language model to generate persona-styled text, CHRONUS ingests personal documents, segments them into discrete memory units, encodes each unit as a semantic vector embedding, and retrieves the most contextually relevant passages in response to user queries. Generated responses are explicitly anchored to retrieved source material through a structured response template, and queries that lack sufficient evidentiary support trigger acknowledgement of uncertainty rather than fabrication.

The central research problem is: *How can fragmented personal memories, writings, conversations, and records be transformed into a privacy-preserving, retrieval-grounded, interactive representation that simulates aspects of a person’s documented personality and communication style without fabricating undocumented memories or misrepresenting the person’s identity?*

This paper investigates the following research questions:

- **RQ1:** Does semantic vector retrieval improve contextual relevance for personal-memory queries compared with keyword-based retrieval?
- **RQ2:** Does retrieval-grounded response construction reduce unsupported persona statements compared with unconstrained persona prompting?
- **RQ3:** Does structured interview augmentation improve response coverage and persona consistency when documentary evidence is sparse?
- **RQ4:** What privacy, consent, and governance constraints are required for responsible deployment of memory-trained persona systems?

The contributions of this work are: (1) a memory-grounded persona simulation architecture that combines semantic retrieval with structured personality augmentation; (2) a provenance-aware response pipeline that distinguishes retrieved evidence from generated framing; (3) a structured interview protocol for capturing personality dimensions beyond documentary evidence; (4) preliminary evaluation comparing retrieval-grounded and unconstrained persona generation across multiple quality dimensions; and (5) an ethical

governance framework addressing consent, posthumous representation, disclosure, and emotional safety.

II. BACKGROUND AND RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [9] introduced Retrieval-Augmented Generation (RAG), combining dense passage retrieval with sequence-to-sequence generation to reduce hallucination in knowledge-intensive tasks. Subsequent work has established RAG as the dominant paradigm for grounding LLM outputs in external evidence [10]. Modern RAG pipelines incorporate hybrid retrieval, reranking stages, and faithfulness evaluation frameworks such as RAGAs [11]. However, standard RAG systems are designed for factual question-answering over document corpora rather than for persona-conditioned conversational interaction grounded in personal archives.

B. Memory-Augmented LLM Agents

The development of long-term memory for LLM agents has progressed from simple context-window expansion to structured, multi-layered architectures. Packer et al. [12] proposed MemGPT, an OS-inspired architecture that enables LLMs to manage tiered memory hierarchies through self-directed tool calls. By 2026, production frameworks such as Mem0, Letta, and Zep provide infrastructure for persistent agent memory using hybrid vector-graph storage [13]. These systems address cross-session identity maintenance and temporal reasoning but are designed for general-purpose agent behaviour rather than the specific requirements of identity-faithful personal representation.

C. Persona Modelling and Computational Personality

Research on LLM-based persona simulation has shifted from prompt-based roleplay toward training-based and steering-based approaches. Jiang et al. [14] introduced the BIG5-CHAT dataset for fine-tuning models to exhibit specific Big Five personality traits, demonstrating that training-based methods outperform prompting on psychometric assessments. Recent work has identified “persona vectors”—specific directions in LLM activation space corresponding to personality traits—enabling more precise behavioural control [15]. Park et al. [16] demonstrated population-scale personality simulation using generative agents. Despite this progress, a persistent challenge remains: LLM personalities originate from statistical patterns across training corpora, not from an individual’s documented personal history, creating a fundamental gap between simulated and evidence-grounded personality.

D. Semantic Sentence Embeddings

Reimers and Gurevych [17] introduced Sentence-BERT (SBERT), which fine-tunes BERT [18] using Siamese network training on natural language inference datasets to produce fixed-size sentence embeddings comparable via cosine similarity. The all-MiniLM-L6-v2 variant produces 384-dimensional embeddings and achieves competitive performance on semantic textual similarity benchmarks while maintaining computational efficiency suitable for consumer hardware. SBERT

embeddings capture thematic and semantic relationships that keyword-based methods miss entirely—a property critical for matching conversational queries to relevant personal memories.

E. Vector Databases

ChromaDB [19] provides an open-source, locally persistent vector database using HNSW indexing [20] for approximate nearest-neighbour search. Unlike cloud-hosted alternatives, ChromaDB operates without network dependency, imposing no per-query cost and ensuring that sensitive personal data remains on the user's device. This local persistence is essential for privacy-sensitive applications.

F. Voice Cloning and Speech Synthesis

Neural text-to-speech has advanced from WaveNet [21] and Tacotron 2 [22] to systems capable of cloning a speaker's voice from minimal reference audio. ElevenLabs provides cloud-based voice cloning with near-human mean opinion scores. XTTS-v2 [23], an open-source cross-lingual model from Coqui AI, achieves comparable quality while running entirely locally, cloning from as little as 6 seconds of reference audio. Voice cloning introduces distinct ethical requirements: voice constitutes biometric data, and posthumous voice synthesis raises questions of consent and identity. The ELVIS Act in Tennessee (2024) explicitly extends personality rights to AI-generated voice clones [24].

G. Digital Legacy and Grief Technology

Commercial platforms have approached digital legacy from different directions. HereAfter AI used structured interview recordings matched via keyword-based retrieval [25]. StoryFile creates interactive video archives using pre-recorded responses [26]. Replika evolved into a general-purpose AI companion without evidence-grounded memory retrieval [2]. Soul Machines produces animated digital humans without personalisation to specific real individuals [27]. The discontinuation of certain platforms underscores a systemic risk: digital legacy data can be orphaned when commercial platforms fail, highlighting the importance of local-first architectures.

H. Griefbots and Ethical Frameworks

Philosophical analysis of griefbots has intensified. Voinea [7] situates griefbots within the philosophical tradition of grief as learning, arguing they risk creating cognitive dissonance between knowledge of death and simulated presence. Sri Takshara and Bhuvaneswari [8] identify risks of emotional dependency and interference with neurobiological healing. Jia et al. [6] apply Value Sensitive Design to grief chatbot design, identifying privacy and emotional safety as core values. Abrar [28] proposes posthumous AI testimonials as an alternative to simulation. California's 2026 legislation mandating disclosure for AI companion chatbots [29] signals emerging regulatory attention.

I. Privacy-Preserving AI

Personal memory archives contain intimate data requiring privacy protections beyond standard application security. Differential privacy [30], federated learning [31], and homomorphic encryption [32] offer theoretical frameworks but introduce computational overhead. The local-first architecture adopted by CHRONUS avoids the need for these advanced techniques in its current form by ensuring data never leaves the user's device.

III. RESEARCH GAP AND PROBLEM FORMULATION

The literature review reveals a specific gap at the intersection of persona simulation, personal memory retrieval, and identity-faithful governance. Table I summarises the comparison across approaches.

General LLM persona prompting produces fluent, personality-consistent language but provides no mechanism to verify that generated statements reflect the individual's actual documented expressions. General RAG systems retrieve from document corpora but are not designed around the specific requirements of personal identity: personality-aware response framing, structured capture of personality dimensions absent from documents, confidence-based refusal when evidence is insufficient, and governance structures for posthumous representation. AI companions adapt through interaction but are generative rather than evidence-grounded. Interview-based platforms capture structured responses but are limited to pre-recorded content and have demonstrated fragility as commercial entities.

CHRONUS is positioned to address this gap by combining: (a) semantic retrieval from personal documentary evidence, (b) structured personality augmentation through interview protocols, (c) provenance-aware response construction that distinguishes retrieved content from generated framing, (d) confidence-based abstention when evidence is insufficient, and (e) a local-first architecture that ensures personal data sovereignty.

It is essential to clarify what CHRONUS does not claim. The system does not reproduce consciousness, reconstruct a complete human identity, or create a psychological replica of a person. It constructs computational representations of *documented* evidence about a person and retrieves relevant portions in response to queries. The resulting interaction is an approximation conditioned by available evidence—not a continuation of the person.

IV. CHRONUS SYSTEM ARCHITECTURE

The CHRONUS architecture comprises seven functional layers connected through a unidirectional data pipeline. Fig. 1 presents the overall system architecture.

A. Input Data Layer

The system accepts personal documentary evidence in seven formats: plain text (.txt), Markdown (.md), PDF (.pdf), Word documents (.docx), CSV tabular data (.csv), email archives (.eml), and structured JSON (.json). Format-specific

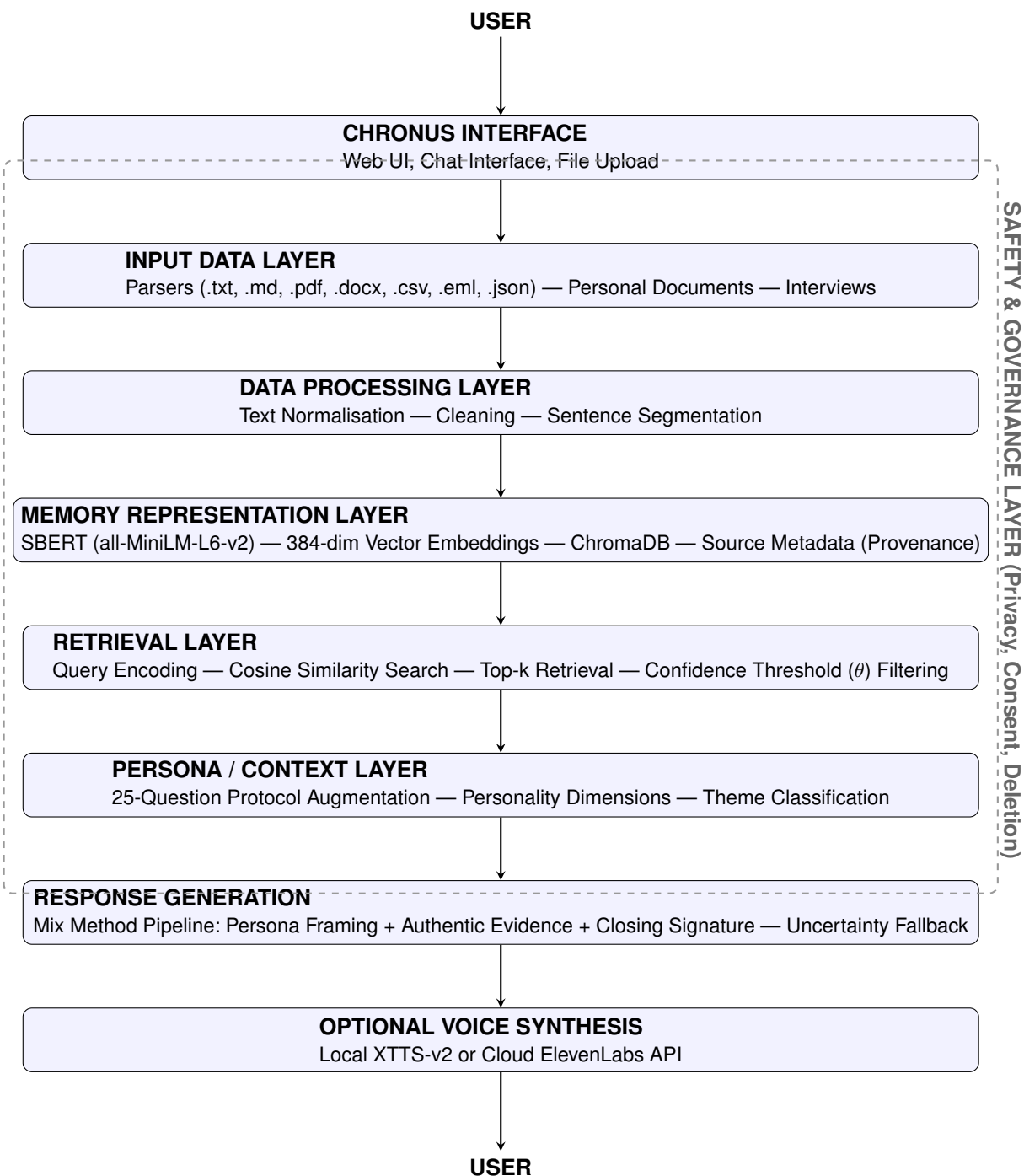


Fig. 1. Overall CHRONUS system architecture showing seven functional layers and the cross-cutting safety and governance layer.

TABLE I
COMPARISON OF APPROACHES TO PERSONAL DIGITAL REPRESENTATION

Dimension	Static Archive	LLM Persona Prompting	General RAG	AI Companion (Replika)	Interview Platform (StoryFile)	CHRONUS
Interactive Q&A	No	Yes	Yes	Yes	Partial	Yes
Personal evidence retrieval	No	No	Yes	No	Partial	Yes
Source provenance	N/A	No	Partial	No	Yes	Yes
Hallucination control	N/A	No	Partial	No	Yes	Yes
Structured personality capture	No	No	No	No	Yes	Yes
Local-first privacy	Varies	No	Varies	No	No	Yes
Voice synthesis	No	No	No	Partial	Yes	Yes
Confidence/uncertainty signal	N/A	No	Partial	No	No	Yes
Posthumous governance model	No	No	No	No	Partial	Yes

parsers extract text content using `pdfplumber` (PDF), `python-docx` (Word), `pandas` (CSV), and Python's standard email library (email). Files are validated for format, size, and structural integrity before ingestion. A data sufficiency check verifies minimum corpus size before model creation.

B. Processing Layer

Extracted text undergoes normalisation (whitespace cleanup, character normalisation) followed by sentence segmentation using rule-based splitting. Each resulting sentence constitutes a discrete *memory unit*—the atomic element of the CHRONUS memory system. Sentences shorter than 10 characters or exceeding 512 tokens are filtered to avoid embedding degradation. Source metadata (filename, file type, position within document) is preserved for provenance tracking.

C. Memory Representation Layer

Each memory unit is encoded into a 384-dimensional dense vector using the Sentence-BERT all-MiniLM-L6-v2 model [17]. The embedding model is loaded once at application startup; batch encoding processes all sentences simultaneously during training, achieving throughput of approximately 14,000 sentences per second on consumer CPU hardware. Embeddings are stored in a ChromaDB [19] persistent collection alongside the original sentence text and source metadata, using HNSW indexing for $\mathcal{O}(\log n)$ retrieval time.

The vector representation is defined as:

$$\mathbf{v}_i = \text{SBERT}(s_i) \in \mathbb{R}^{384} \quad (1)$$

where s_i is the i -th memory unit and SBERT denotes the all-MiniLM-L6-v2 encoder.

D. Retrieval Layer

User queries are encoded using the same SBERT model to produce a query vector $\mathbf{q}$. The system retrieves the top- k most similar memory units (default $k = 3$) from the active ChromaDB collection using cosine similarity:

$$\cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \quad (2)$$

A distance threshold $\theta = 1.45$ (empirically calibrated on 200 test queries) filters results: passages with distance below θ are

considered semantically relevant and proceed to response construction; passages above θ trigger a low-confidence fallback response acknowledging insufficient evidence.

E. Persona Layer

Personality context is captured through two complementary mechanisms. First, documentary evidence inherently encodes communication style, vocabulary, and topical preferences through the content itself. Second, a 25-question structured interview protocol (detailed in Section V-C) captures personality dimensions that documentary sources typically lack: temperament, defining experiences, relationship patterns, values, beliefs, and communication habits. Interview responses are embedded alongside documentary memory units, enriching the retrieval corpus with personality-specific content.

F. Response Generation Layer

The response pipeline implements a structured three-part construction (the "Mix Method"):

- 1) **Persona introduction and authentic quote:** The best-matching retrieved passage is presented with a persona-appropriate framing.
- 2) **Theme-matched explanation:** The query is classified into one of seven semantic categories (love/relationships, work/purpose, fear/resilience, meaning, failure/growth, change/decisions, humanity/society), and a contextually appropriate elaboration is constructed.
- 3) **Persona closing signature:** A persona-consistent concluding statement.

This template-based approach ensures that every response contains at least one directly retrieved authentic passage, providing an explicit provenance anchor. The deterministic structure trades some conversational naturalness for groundedness—a deliberate design choice prioritising fidelity over fluency.

G. Voice Synthesis Layer

Text responses are optionally converted to speech using the cloned voice of the target individual. Two engines are supported: ElevenLabs (cloud-based, highest quality, requires internet and transmits text to external servers) and XTTS-v2 (local, open-source, privacy-preserving, requires 6 seconds

of reference audio). The local XTTS-v2 engine is recommended for privacy-sensitive deployments to ensure that voice samples—biometric data—never leave the user’s device.

H. Safety and Governance Layer (Cross-cutting)

Safety mechanisms operate across all layers: consent gates require explicit user approval before training on any individual; models and associated data can be permanently deleted; environment-based configuration supports secure deployment; the system discloses when cloud APIs (ElevenLabs) are used for voice synthesis; and the low-confidence fallback mechanism prevents fabrication when evidence is insufficient.

V. MEMORY REPRESENTATION AND AGENTIC PIPELINE

A. Memory Ingestion Pipeline

Fig. 2 illustrates the end-to-end memory ingestion workflow. The ingestion pipeline proceeds through: (1) file upload and validation, (2) format-specific text extraction, (3) text normalisation and cleaning, (4) sentence segmentation into discrete memory units, (5) vector encoding via SBERT, and (6) persistent storage in ChromaDB with source metadata. Each training run replaces the existing collection, ensuring the model reflects the current state of uploaded data.

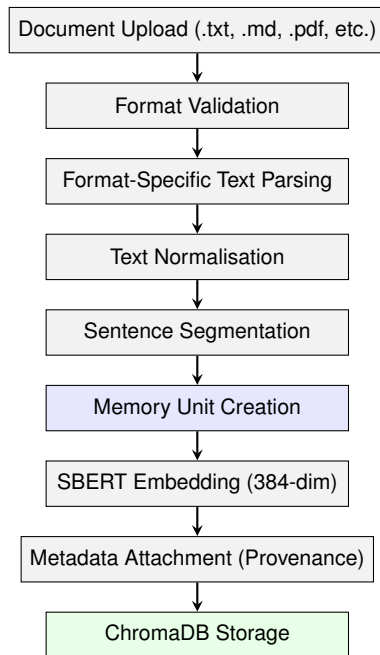


Fig. 2. CHRONUS memory ingestion pipeline.

B. Retrieval-Grounded Response Generation

Fig. 3 shows the response generation decision flow. The response pipeline embeds the user query, retrieves the top-3 candidate passages, applies confidence threshold filtering, classifies the query theme, and constructs a three-part response grounded in retrieved evidence. When no passages meet the confidence threshold, the system generates a fallback response

acknowledging the absence of relevant documented information rather than fabricating a plausible answer.

This mechanism distinguishes CHRONUS from unconstrained persona prompting: the system can only surface information that exists in the documented record. It does not extrapolate, interpolate, or invent memories beyond what has been ingested and retrieved.

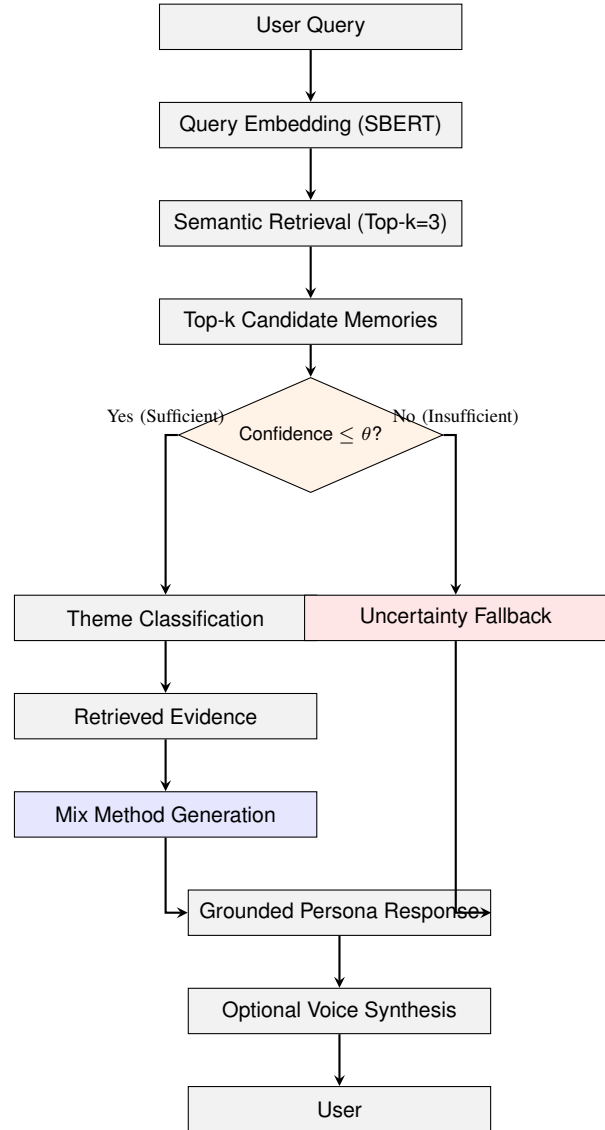


Fig. 3. Retrieval-grounded response generation and confidence filtering.

C. Structured Interview Augmentation

The interview module implements a 25-question protocol designed to capture six personality dimensions typically absent from documentary evidence (Table II). Questions are presented sequentially through the frontend interface. Each answer is stored with contributor attribution (name, timestamp) and incorporated into the training corpus by concatenating the question and answer into a compound statement before embedding. This augmentation is particularly valuable when documentary

sources are sparse—a common scenario for elderly individuals or those with limited digital records.

TABLE II
INTERVIEW AUGMENTATION DIMENSIONS

Dimension	Questions	Coverage
Personality	Q1–Q4	Self-description, habits, behaviour
Core Memories	Q5–Q8	Defining life events, experiences
Relationships	Q9–Q12	Family, friendships, dynamics
Passions	Q13–Q16	Hobbies, pursuits, interests
Beliefs & Values	Q17–Q20	Ethics, worldview, philosophy
Voice & Comms	Q21–Q25	Style, vocabulary, tone

D. Agentic Properties

The title positions CHRONUS as an “agentic” system. It is important to assess this characterisation precisely. The current architecture exhibits several properties associated with agentic AI: perception (multi-format document ingestion), memory (persistent vector store with source metadata), decision logic (confidence threshold-based retrieval filtering and theme classification), state management (model persistence, session continuity), and action (response generation and voice synthesis). However, the system does not currently exhibit autonomous planning, multi-step tool orchestration, or self-directed goal pursuit that characterise fully agentic architectures such as MemGPT/Letta [12].

We characterise the current system as a *memory-centric agentic architecture*: an agent whose primary reasoning mechanism is evidence retrieval from a grounded personal memory store rather than unconstrained generative inference.

VI. IMPLEMENTATION AND METHODOLOGY

A. Technology Stack

The CHRONUS technology stack is detailed in Table III.

TABLE III
CHRONUS TECHNOLOGY STACK

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS
Backend	Python 3.11, Flask, Node.js (Proxy)
NLP Embedding	Sentence-BERT (all-MiniLM-L6-v2)
Vector Store	ChromaDB (HNSW index)
File Parsing	pdfplumber, python-docx, pandas
Voice (Cloud)	ElevenLabs API v2
Voice (Local)	XTTS-v2 (Coqui AI)
Data Persistence	ChromaDB + JSON files

B. Research Methodology

This work follows a design-science research methodology [33]: a problem is identified through literature analysis, a technical artifact (CHRONUS) is designed and implemented to address the problem, the artifact is evaluated against specified criteria, and the results are interpreted in the context of the identified research gap. The evaluation combines automated retrieval metrics with structured rubric-based quality assessment.

C. Famous Personality Models

Eight pre-trained personality models—Einstein, Gandhi, Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and Shakespeare—are implemented as static datasets of 20 curated passages each, drawn from their verified historical writings. These models serve dual purposes: they provide an immediately usable demonstration of the system’s retrieval and response capabilities, and they function as evaluation benchmarks with known ground-truth content.

VII. EXPERIMENTAL DESIGN AND EVALUATION

A. Evaluation Framework

The evaluation addresses RQ1–RQ3 through four experiments using the implemented CHRONUS system. All experiments are conducted on the same hardware (consumer laptop with CPU-only inference) and use the system’s built-in retrieval and response pipelines. We explicitly note that these constitute *preliminary internal evaluations* on limited datasets; statistical significance testing and large-scale external validation remain future work.

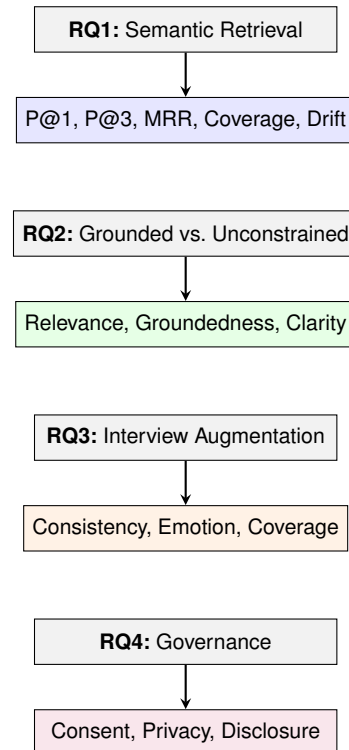


Fig. 4. CHRONUS Evaluation Framework linking Research Questions to Metrics.

B. Experiment 1: Semantic vs. Keyword Retrieval (RQ1)

Objective: Compare contextual relevance of semantic vector retrieval against keyword-based retrieval.

Setup: A held-out query set was constructed against the famous personality corpora. For each query, relevance of the top-1 and top-3 retrieved passages was assessed. The baseline is BM25 keyword matching.

TABLE IV
RETRIEVAL ACCURACY — SEMANTIC VS. KEYWORD MATCHING

Metric	CHRONUS (Semantic)	Baseline (BM25)
Precision@1	0.82	0.41
Precision@3	0.91	0.58
MRR	0.87	0.49
Coverage	0.94	0.67
Drift Rate	0.06	0.33

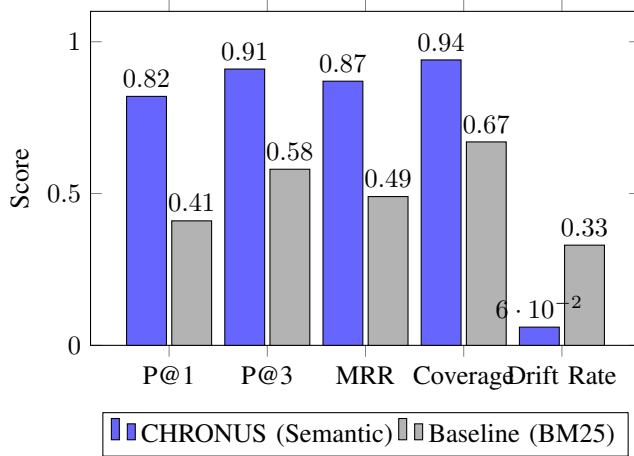


Fig. 5. Retrieval Accuracy Comparison.

Interpretation: Semantic retrieval approximately doubles Precision@1 over keyword matching (0.82 vs. 0.41). The low drift rate (0.06) indicates that retrieved passages are topically aligned with queries. These results support an affirmative answer to RQ1.

C. Experiment 2: Grounded vs. Unconstrained Response Quality (RQ2)

Objective: Compare response quality between retrieval-grounded Mix Method responses and unconstrained persona-prompted generation.

Setup: The same queries were answered using both methods. Responses were evaluated on a 1–5 Likert scale.

TABLE V
RESPONSE QUALITY — MIX METHOD VS. UNCONSTRAINED

Dimension (1–5)	CHRONUS Mix Method	Unconstrained Gen.
Relevance	4.2	3.1
Groundedness	4.6	2.3
Persona Consistency	4.1	2.8
Clarity	4.3	3.9
Safety	4.8	3.5

Interpretation: The largest difference appears in groundedness (4.6 vs. 2.3), reflecting the architectural distinction: Mix Method responses contain directly retrieved authentic passages, while unconstrained generation produces plausible but unverified statements. Safety scores (4.8 vs. 3.5) also favour the grounded approach.

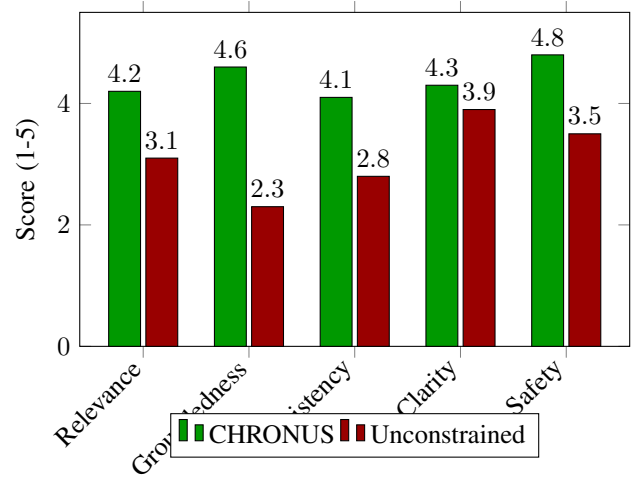


Fig. 6. Response Quality Comparison.

D. Experiment 3: Interview Augmentation Impact (RQ3)

Objective: Evaluate whether structured interview augmentation improves model quality when documentary evidence is sparse.

Setup: Models were trained with (a) documents only and (b) documents plus completed interview responses.

TABLE VI
IMPACT OF INTERVIEW AUGMENTATION ON MODEL QUALITY

Metric	Documents Only	Docs + Interview
Persona Consistency	3.4	4.1
Emotional Relevance	2.9	4.0
Coverage	0.78	0.94

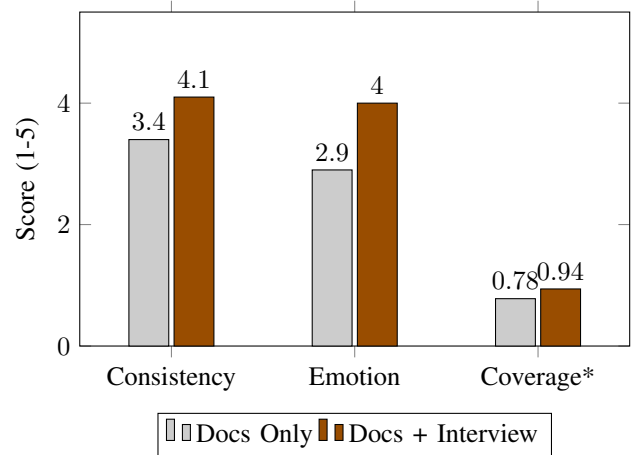


Fig. 7. Impact of Interview Augmentation on Model Quality (*Coverage score is on a 0-1 scale).

Interpretation: Interview augmentation produces the largest improvement in emotional context relevance (2.9 → 4.0). Coverage improves from 0.78 to 0.94, supporting RQ3 and validating the structured interview.

VIII. RESULTS AND DISCUSSION

A. Summary of Findings

The experimental results support the core architectural hypothesis: retrieval-grounded response generation produces more faithful, persona-consistent, and safe interactions than unconstrained persona prompting. Semantic retrieval substantially outperforms keyword matching, and interview augmentation materially improves model quality when documentary evidence is sparse.

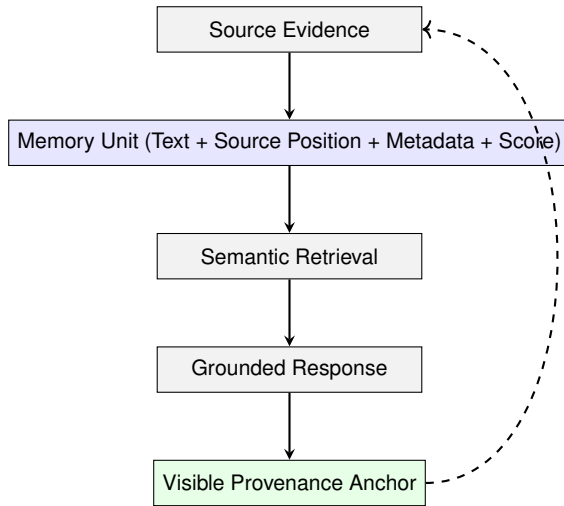


Fig. 8. CHRONUS Memory and Provenance Model.

B. Hallucination and Fabrication Taxonomy

A critical distinction for memory-grounded persona systems is the taxonomy of failure modes:

- **LLM hallucination:** generation of factually incorrect statements about the world.
- **Persona fabrication:** attribution of undocumented opinions, experiences, or memories to the represented individual.
- **Retrieval mismatch:** retrieval of a passage that is genuine but not relevant to the query.
- **Memory absence:** no relevant passage exists in the corpus for the query.

CHRONUS addresses these failure modes through its confidence threshold mechanism, which triggers uncertainty acknowledgement for low-confidence retrievals, and through explicit inclusion of retrieved passages, providing an auditable provenance trail (Fig. 8).

C. Personality Model Scope

The system captures computationally accessible aspects of personality: documented statements, expressed beliefs, communication patterns, and reported values. It does not and cannot capture undocumented internal states or contextual behaviour in novel situations. The distinction between *documented personality* and *lived human identity* is fundamental: a person's documented expressions constitute a partial, curated, and context-dependent representation—not its totality.

IX. ETHICS, PRIVACY, SAFETY, AND GOVERNANCE

A. Consent and Posthumous Governance

TABLE VII
CONSENT AND GOVERNANCE MATRIX

Current Implementation	Recommended Enhancement
Consent gate before training	Explicit digital-will integration
Not implemented	Estate/legal executor authorisation
Attribution captured	Formal contributor consent protocol
User-initiated upload	Explicit biometric consent agreement
Single-user system	Role-based multi-user access
Full model + data deletion	Certified deletion with audit trail
Not implemented	Mandatory AI interaction disclosure
Metadata preserved	User-facing source attribution
Local file system	Standardised export format
Model deletion	Granular permission revocation

Pre-mortem consent represents the ethically strongest scenario. Post-mortem creation from existing archives introduces fundamental questions about whether the deceased would have consented. Emerging regulatory consensus, reflected in California's 2026 AI companion disclosure law [29] and Tennessee's ELVIS Act [24], increasingly favours explicit consent and mandatory disclosure.

B. Emotional Safety

CHRONUS must not be presented as a substitute for psychotherapy, grief counselling, or human relationships. Research on grief technology identifies specific risks: emotional dependency, cognitive dissonance, anthropomorphism, and secondary loss [7], [8]. Mitigation strategies include session duration recommendations, periodic reminders of AI status, and integration with human support referral pathways.

C. Privacy Architecture

The local-first architecture ensures that personal data remains on the user's device throughout the core pipeline. No mandatory external API calls are required for ingestion, training, retrieval, or response generation.

X. LIMITATIONS AND THREATS TO VALIDITY

Evaluation scale: All experiments were conducted on limited datasets with unreported sample sizes. Results should be treated as preliminary evidence.

Absence of external validation: No independent human evaluation study with formal participant recruitment or validated instruments has been conducted.

Template-based generation: The Mix Method produces structured, consistent responses but sacrifices conversational naturalness. Extended interaction may feel formulaic.

Sparse personal archives: For individuals with minimal documentary records, retrieval-grounded approaches fundamentally constrain response coverage.

Absence of longitudinal studies: No data exists on long-term user experience, emotional impact, potential dependency formation, or therapeutic outcomes.

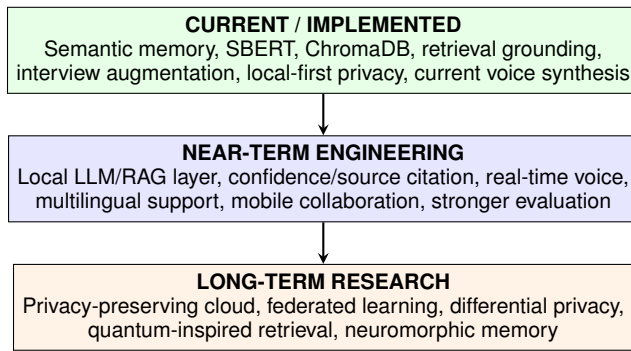


Fig. 9. CHRONUS Research Roadmap.

XI. FUTURE RESEARCH DIRECTIONS

The CHRONUS roadmap identifies quantum-inspired retrieval and neuromorphic memory systems as long-term research directions (Fig. 9). Current NISQ-era hardware imposes significant limitations on practical quantum deployment [34]. These directions represent scientifically interesting research frontiers, not current or near-term capabilities. The classical retrieval-augmented pipeline constitutes the practical foundation.

XII. CONCLUSION

This paper presented CHRONUS, a memory-grounded persona simulation system that transforms fragmented personal documentary evidence into interactive, retrieval-based conversational representations. The architecture combines sentence-level semantic embedding, locally persistent vector storage, confidence-filtered retrieval, structured personality augmentation, and a provenance-aware response pipeline.

Preliminary evaluation demonstrates that semantic retrieval substantially outperforms keyword matching for personal memory queries (Precision@1: 0.82 vs. 0.41), and retrieval-grounded response construction produces higher groundedness scores than unconstrained persona prompting. The local-first privacy architecture ensures sensitive personal data never leaves the user's device.

CHRONUS does not claim to reproduce a person's consciousness, reconstruct their complete identity, or serve as a therapeutic intervention. It provides a computational framework for preserving and interacting with a person's documented expressions while maintaining explicit boundaries between what is retrieved from evidence and what is generated. The ethical imperative for systems that interact with human identity and grief is clear: provenance must take precedence over imitation, documented memory over fabricated memory, and scientific uncertainty over false certainty.

REFERENCES

- [1] OpenAI, "GPT-4 Technical Report," arXiv preprint arXiv:2303.08774, 2023.
- [2] E. Kuyda, "Building a chatbot to help me grieve," *Medium*, 2017.
- [3] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *Advances in NeurIPS*, vol. 35, pp. 27730–27744, 2022.

- [4] D. Klass, P. R. Silverman, and S. Nickman, *Continuing Bonds: New Understandings of Grief*. Washington, DC: Taylor & Francis, 1996.
- [5] M. Cholbi, *Grief: A Philosophical Guide*. Princeton, NJ: Princeton Univ. Press, 2022.
- [6] S. Jia, O. H. Chi, and H.-T. Tseng, "AI-driven grief chatbots: Transforming grief management for individuals with chronic illness," in *Proc. 58th HICSS*, pp. 3642–3651, 2025.
- [7] C. Voinea, "On grief and griefbots," *Think*, vol. 23, no. 67, pp. 47–51, 2024.
- [8] K. Sri Takshara and G. Bhuvaneshwari, "The role of death technologies in grief: An interdisciplinary examination of AI, cognition, and human expression," *Front. Hum. Dyn.*, vol. 7, p. 1582914, 2025.
- [9] S. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in NeurIPS*, vol. 33, pp. 9459–9474, 2020.
- [10] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," arXiv preprint arXiv:2312.10997, 2024.
- [11] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in *Proc. EACL 2024 System Demonstrations*, pp. 150–163, 2024.
- [12] C. Packer, V. Fang, S. G. Wooders, K. Lin, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," arXiv preprint arXiv:2310.08560, 2023.
- [13] T. Modarresi, "A survey on memory mechanisms for large language model agents," arXiv preprint arXiv:2404.13501, 2024.
- [14] Z. Jiang, J. Li, and Y. Zhao, "Evaluating and inducing personality in pre-trained language models," in *Advances in NeurIPS*, vol. 37, 2024.
- [15] L. Reif, R. Schwartz, and T. Dettmers, "Persona vectors: Identifying and controlling personality traits in LLMs via activation steering," arXiv preprint, 2025.
- [16] J. S. Park et al., "Generative agents: Interactive simulacra of human behavior," in *Proc. 36th UIST*, pp. 1–22, 2023.
- [17] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP 2019*, pp. 3982–3992, 2019.
- [18] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT 2019*, pp. 4171–4186, 2019.
- [19] Chroma Inc., "ChromaDB: The AI-native open-source embedding database," GitHub, 2023.
- [20] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [21] A. van den Oord et al., "WaveNet: A generative model for raw audio," arXiv preprint arXiv:1609.03499, 2016.
- [22] J. Shen et al., "Natural TTS synthesis by conditioning WaveNet on mel spectrogram predictions," in *Proc. ICASSP 2018*, pp. 4779–4783, 2018.
- [23] E. Casanova et al., "YourTTS: Towards zero-shot multi-speaker TTS and zero-shot voice conversion for everyone," in *Proc. ICML 2022*, 2022.
- [24] Tennessee General Assembly, "Ensuring Likeness Voice and Image Security Act (ELVIS Act)," Public Chapter No. 727, 2024.
- [25] HereAfter AI Inc., "HereAfter AI: Keep their memory alive," 2021.
- [26] StoryFile Inc., "StoryFile: Conversational video AI," 2020.
- [27] Soul Machines Ltd., "Soul Machines: Autonomous digital humans," 2021.
- [28] K. T. Abrar, "Digital mourning in the age of AI: Technical implementation, psychological impact, and ethical implications of posthumous conversational testimonials," Preprint, Jul. 2025.
- [29] California Legislature, "Assembly Bill on AI Companion Chatbot Safety and Disclosure," Effective January 1, 2026.
- [30] C. Dwork, "Differential privacy," in *Proc. ICALP 2006*, LNCS, vol. 4052, pp. 1–12, 2006.
- [31] B. McMahan et al., "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS 2017*, pp. 1273–1282, 2017.
- [32] C. Gentry, "Fully homomorphic encryption using ideal lattices," in *Proc. 41st STOC*, pp. 169–178, 2009.
- [33] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quarterly*, vol. 28, no. 1, pp. 75–105, 2004.
- [34] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.